COURSE-BASED PROJECT DOCUMENTATION | DOC-08

Syntax Parser (GNU Bison LALR(1))

Context-Free Grammars, Shift-Reduce Automata, & Conflict Profile

1. Syntax Analysis & Context-Free Grammars

Syntax analysis groups the stream of tokens into grammatical structures defined by a Context-Free Grammar (CFG). GNU Bison generates a deterministic bottom-up LALR(1) (Look-Ahead Left-to-Right with 1 token lookahead) state machine. The parser maintains a parse stack and consults Action and Goto tables to perform Shift and Reduce operations.

2. Shift-Reduce Parser Operations

- ? Shift: Pushes the current lookahead token onto the stack and advances the input stream.
- ? Reduce: Replaces a sequence of grammar symbols on top of the stack matching rule A -> alpha with nonterminal A.
- ? Accept: Start symbol compilationUnit is reduced on EOF ($end); syntax is valid.
- ? Error: Action table entry is empty for the (State, Token) pair; triggers yyerror().

3. Operator Precedence & Associativity Table

Precedence Level	Associativity	Operators / Directives
1. Dangling Else	%nonassoc	LOWER_THAN_ELSE, ELSE
2. Lambda Arrow	%right	ARROW (->)
3. Assignments	%right	=, +=, -=, *=, /=, %=, &=, ^=, |=, <<=, ...
4. Ternary	%right	?, :
5. Logical OR / AND	%left	||, &&
6. Equality / Relational	%left	==, !=, <, >, <=, >=, instanceof
7. Shifts / Additive	%left	<<, >>, >>>, +, -
8. Multiplicative	%left	*, /, %
9. Unary / Prefix	%right	++, --, +, -, ~, !, (cast)

4. Bison Conflict Profile (285 S/R, 34 R/R)

Bison reports 285 shift/reduce and 34 reduce/reduce conflicts in java_parser.output. These arise from contextual keywords (record, when, yield, var) that can serve as either keywords or identifiers, and Java's statement grammar. Bison's default shift-over-reduce resolution perfectly matches Java's longest-match semantics.